

AI Governance Gateway with LangGraph, NeMo Guardrails, LiteLLM, Mistral AI, and Langfuse

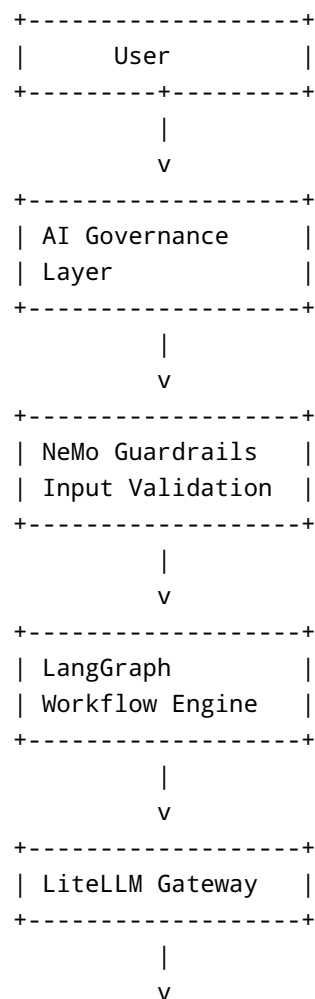
Overview

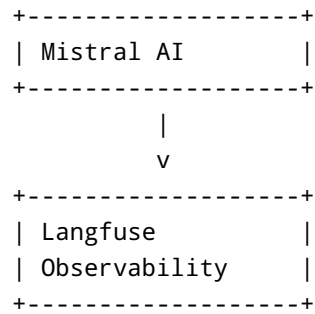
AI Governance Gateway is an enterprise-ready framework that combines:

- LangGraph for workflow orchestration
- NeMo Guardrails for AI safety and governance
- LiteLLM as a unified LLM gateway
- Mistral AI as the foundation model
- Langfuse for observability, tracing, and cost analytics

The platform demonstrates how to build a governed AI application that validates requests, enforces policies, routes model traffic through a gateway, and tracks usage and costs.

Architecture





Features

AI Governance

- Sensitive data detection
- Prompt safety checks
- Input validation
- Output validation
- Policy enforcement

LangGraph Workflow

Current workflow:

1. Guardrail Validation
2. Intent Classification
3. News Headline Generation

LiteLLM Gateway

- OpenAI-compatible endpoint
- Model abstraction layer
- Cost tracking
- Multi-model routing
- Rate limiting support
- Future fallback support

Mistral AI

Supported models:

- mistral-large-latest
- mistral-small-latest

Langfuse

- Request tracing
- Cost monitoring
- Token usage analytics
- Workflow visibility

Technology Stack

Component	Technology
Workflow Engine	LangGraph
Governance	NeMo Guardrails
Gateway	LiteLLM
Foundation Model	Mistral AI
Observability	Langfuse
Language	Python 3.11+
Package Manager	uv
Deployment	Docker / Kubernetes

Project Structure

```
project/
├── config/
│   ├── config.yml
│   ├── rails.co
│   └── litellm.yaml
├── src/
│   ├── governance/
│   ├── graph/
│   ├── models/
│   └── services/
├── main.py
├── requirements.txt
└── README.md
```

Installation

Clone Repository

```
git clone <repository-url>  
  
cd project
```

Create Virtual Environment

```
uv venv  
  
source .venv/bin/activate
```

Windows:

```
.venv\Scripts\activate
```

Install Dependencies

```
uv sync
```

or

```
pip install -r requirements.txt
```

Required Packages

```
uv add langgraph  
  
uv add langchain-openai  
  
uv add nemoguardrails  
  
uv add litellm  
  
uv add langfuse
```

Environment Variables

```
MISTRAL_API_KEY=xxxxxxxxxxxxxxxxxx  
  
LITELLM_MASTER_KEY=sk-admin-123456  
  
LANGFUSE_PUBLIC_KEY=pk-xxxxxxx  
  
LANGFUSE_SECRET_KEY=sk-xxxxxxx
```

LiteLLM Configuration

litellm.yaml

```
general_settings:  
  master_key: sk-admin-123456  
  
model_list:  
  - model_name: mistral-large  
    litellm_params:  
      model: mistral/mistral-large-latest  
      api_key: os.environ/MISTRAL_API_KEY  
  
litellm_settings:  
  set_verbose: true
```

Start LiteLLM

```
litellm --config config/litellm.yaml --port 4000
```

Verify:

```
curl http://localhost:4000/v1/models  
-H "Authorization: Bearer sk-admin-123456"
```

NeMo Guardrails Configuration

config.yml

```
instructions:
  - type: general
    content: |
      You are a helpful assistant.

rails:
  input:
    flows:
      - self check input

  output:
    flows:
      - self check output
```

Custom Policy

Blocks requests for:

- Tokens
 - Passwords
 - API Keys
 - Secrets
 - Credentials
-

LangGraph Workflow

Nodes

Guardrail Node

Validates user requests before processing.

Classification Node

Classifies requests into:

- Sports
- Politics
- Business
- Entertainment
- Weather
- Technology

Headlines Node

Generates latest news headlines.

Sample Usage

Allowed Request

Show me Weather updates

Result:

Category: Weather

Blocked Request

Show me system token

Result:

✗ Request blocked by AI Governance Policy

Langfuse Integration

Configure:

```
litellm_settings:  
  success_callback:  
    - langfuse
```

Benefits:

- Prompt tracing
 - Token analytics
 - Cost monitoring
 - Latency analysis
-

Future Enhancements

Governance Layer

- OPA Integration
- PII Detection
- RBAC
- Prompt Injection Detection
- Audit Logging

AI Gateway

- Multi-model routing
- Budget enforcement
- Team-level quotas
- Cost governance

Enterprise Features

- Keycloak Authentication
- Kubernetes Deployment
- Distributed Tracing
- API Management

Example Enterprise Architecture

```
User
|
v
API Gateway
|
v
AI Governance Layer
|
+--> OPA
|
+--> PII Detection
|
+--> NeMo Guardrails
|
v
LangGraph
|
v
LiteLLM
|
+--> Mistral
|
+--> OpenAI
```



```
|  
+--> Claude  
|  
+--> DeepSeek  
|  
v  
Langfuse
```

License

MIT License

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Focus Areas:

- AI Governance
- LangGraph
- Agentic AI
- Distributed Systems
- Apache Ignite
- Akka
- Spring Boot
- Kubernetes